

Experiment 4: Vacuum Cleaner Problem

Aim

To simulate the Vacuum Cleaner problem using Python.

Objective

To clean all dirty rooms automatically.

Algorithm

1. Check each room.
2. If dirty, clean it.
3. Move to the next room.
4. Repeat until all rooms are clean.

Flowchart

Start

|

Check Room

|

Dirty?

/ \

Yes No

| |

Clean Move

\ /

Next Room

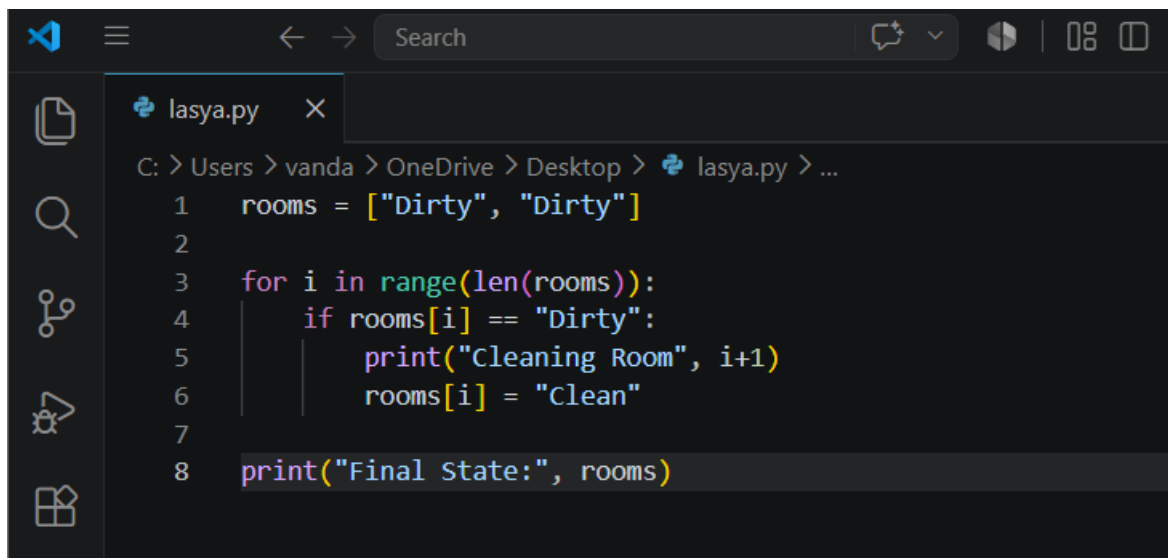
|

All Clean?

|

Stop

Code and Output



```
lasya.py
C: > Users > vanda > OneDrive > Desktop > lasya.py > ...
1  rooms = ["Dirty", "Dirty"]
2
3  for i in range(len(rooms)):
4      if rooms[i] == "Dirty":
5          print("Cleaning Room", i+1)
6          rooms[i] = "Clean"
7
8  print("Final State:", rooms)
```

```
Solution Found
PS C:\Users\vanda\onedrive\desktop> python lasya.py
Cleaning Room 1
Cleaning Room 2
Final State: ['Clean', 'Clean']
```

Result

All rooms were cleaned successfully.

Conclusion

The Vacuum Cleaner agent cleaned every dirty room automatically.